Hannah Estes

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EDUCATION

Ph.D. Computer Science, North Carolina State University — May 2029 (expected)
Advised by Dr. Kathryn T. Stolee, Wolfpack Innovations in Software Engineering Research (WISER) Lab

M.S. Computer Science, North Carolina State University — December 2026 (expected)

B.S. Computer Science, North Carolina State University, Magna Cum Laude — May 2024
Minors: Cognitive Science, Global Leadership & Team Decision Making
NCSU College of Engineering Faculty Senior Scholar Nominee, CSC Ambassador

REFEREED PUBLICATIONS

[1] **Hannah Estes**, Satwika Kancharla, and Kathryn T. Stolee. The Evolution of Software Product Manager Role Expectations in Software Teams. 2026. [Currently Under Review]

[2] **Hannah Estes**, Satwika Kancharla, and Kathryn T. Stolee. Software Product Managers, Software Teams, and the Requirements Between Them. 2026. [Currently Under Review]

[3] Tiva Rocco, **Hannah Estes**, and Sarah Heckman. Comparing What Rule-based and LLM-based Tools Report on Student Team Projects. 2026. [Currently Under Review]

[4] **Hannah Estes** and Kathryn T. Stolee. Who Shapes the Stance? A Stases-Based Analysis of Writer-LLM Negotiation with CoAuthor. Visual Languages and Human Centered Computing (VL/HCC), 2026.

[5] **Hannah Estes**, Katie Hollowell, John Cheek, Shiv Patel, Hanna Reese, and Kathryn T. Stolee. 2025. Advancing Software Product Management Education: Insights from an Industry Survey. In 33rd ACM International Conference on the Foundations of Software Engineering (FSE Companion '25), June 23–28, 2025, Trondheim, Norway. ACM, New York, NY, USA, 11 pages. [DOI]

RESEARCH EXPERIENCE

Graduate Research Assistant — August 2024 - present
WISER Lab, Department of Computer Science, NCSU — Raleigh, NC

- Spearheaded a comprehensive research initiative and designed an IRB-approved protocol to define the software product manager (SPM) role, analyzing survey data from 80 software professionals across 22 companies and 37 job descriptions
- Coordinated a team of five student researchers through the full research pipeline — survey design, data collection, qualitative analysis, and paper writing — resulting in four students earning co-authorship on [5]
- Published and presented a paper at FSE-SEET identifying 21 competencies and 10 key organizational values of the SPM role [5]; completed the written preliminary examination building on this work to propose curriculum enhancements for NCSU CS students
- Led a second paper, submitted to TOSEM, uncovering SPM role expectations across three perspectives — SPMs, software teams, and employers — through surveys of 80 software professionals across 22 companies and analysis of 37 job descriptions; synthesized findings into the SENSE framework characterizing SPM activities and eight friction points across the software development lifecycle **??**
- Applied research findings to inform and update the curriculum of a new SPM course at NCSU, providing students with a competitive advantage in the workforce

Undergraduate Research Assistant under Dr. Kathryn T. Stolee — December 2023 - August 2024

- Laid the groundwork for the SPM research program detailed under *Graduate Research Assistant*, including initial study design and IRB protocol development.

WORKING EXPERIENCE

TITANS Software R&D Intern May 2026 - August 2026
Sandia National Labs Albuquerque, NM

- Utilized large language model (LLM) question-answering techniques to generate and refine hypotheses for open-ended intelligence questions lacking definitive answers, broadening analytical perspectives and enhancing insight quality
- Created knowledge graphs to develop interactive visualizations that map complex relationships among questions, hypotheses, claims, and evidence, in order to improve report clarity, with a goal of aiding intelligence analysts in mitigating cognitive biases during high-stakes decision-making
- Leveraged argumentation schemes to develop LLM-as-judge rubrics for systematically evaluating outputs from multi-agent intelligence systems, enhancing the rigor, consistency, and reliability of hypothesis assessments

Software Engineering Intern June 2024 - August 2024
Pearson Durham, NC

- Consolidated identities and created a seamless ID verification process for in-person and online users
- Developed authentication session transfer capabilities for document verification across multiple devices

Technology Security & Privacy Consulting Intern June 2023 - August 2023
Protiviti Dallas, TX

- Developed an interactive mock online order page utilizing Microsoft Power Apps and Power Pages, showcasing the practical application of digital twins and order management solutions for prospective clients
- Automated tracking of deliverables to support the client's virtualization process

TEACHING EXPERIENCE

CSC 408: Software Product Management, North Carolina State University Fall 2024, Spring 2025, Fall 2026
Teaching Assistant (TA), Instructor: Dr. Kathryn T. Stolee

- Leveraged firsthand experience as one of the first students enrolled in the course to propose updated learning objectives, revised course material, and a redesigned semester project to better align students with industry expectations
- Delivered lectures on qualitative data analysis best practices, guiding students in drawing actionable insights from primary data
- Hosted office hours focused on helping students synthesize findings and develop cohesive, evidence-based narratives for their course projects

INVITED PANELS

Computer Science Grad School Panel NCSU Computer Science November 2026
AI Leadership Summit Student Perspectives NCSU College of Engineering October 2026
Graduate School…should I go? Is it for me? NCSU Engineering Your Experience Workshop February 2025
What is Research? Fontainebleau High School December 2024

SERVICE

Student Volunteer, ACM AI Leadership Summit August 2026

Student Volunteer, International Conference on Software Engineering April 2026

Student Lead & Mentor, Computer Science Mentorship Program August 2025 - present
North Carolina State University

- Lead and coordinate a departmental mentorship program supporting undergraduate CS sophomore students
- Provide one-on-one mentorship on academic planning, career exploration, and professional development to support student retention and success within the department

Founder & Program Lead, PhD Connection Program February 2026 - present
North Carolina State University

- Designed and lead a mentorship initiative pairing incoming Ph.D. students with current students to support transition into graduate school
- Coordinate mentor recruitment, matching, and program communication in collaboration with the Graduate Student Association
- Facilitate early connections among students and with faculty, supporting onboarding and engagement in the months leading into the program

CS Ambassador August 2022 - present
North Carolina State University

- Selected as one of two department representatives at the Business of Software Conference, applying research to highlight the importance of understanding product value and customer delivery
- Delivered talks on computer science, the department, and life after NCSU to groups of 50+ prospective students and parents across four biannual open houses
- Mentored incoming ambassadors on communicating CS-specific concepts to non-technical audiences; facilitated Q&A sessions for prospective and incoming undergraduate students
- Represented the department at the alumni Hall of Fame induction ceremony

Website Coordinator & Communications Liaison, AGEP-NC 2023 - 2024
Alliances for Graduate Education and the Professoriate North Carolina (NSF-funded)

- Supported an NSF-funded program connecting three NC universities (NCSU, NC A&T, UNCC) across eleven STEM doctoral departments to drive systematic change for underrepresented minority doctoral students
- Redesigned and maintained the program website to improve accessibility and usability
- Managed monthly newsletters communicating program progress and resources to stakeholders

MENTORSHIP

Key: ^{u} undergraduate student ^{g} graduate student ^{r} research mentorship

Research Mentees

Satwika Kancharla^{u,r}	Summer 2024, Fall 2024, Spring 2025
Mulikatu Yakuba^{g,r}	Fall 2024
Katja Mathesius^{g,r}	Fall 2024
Katie Hollowell^{u,r}	Spring 2025
John Cheek^{u,r}	Spring 2025
Shiv Patel^{u,r}	Spring 2025
Hanna Reese^{u,r}	Spring 2025

General Mentees

Miriam Bloomfield^{u}	Fall 2025
Nihith Chanapathi^{u}	Fall 2025, Spring 2026
Justin Huang^{u}	Fall 2025
Jacob Garcia^{u}	Spring 2026
Swathi Dwarampudi^{u}	Spring 2026

HONORS, AWARDS, FELLOWSHIPS

AI Leadership Summit Travel Grant ACM, $700 July 2026
VL/HCC Travel Grant NSF, $2000 July 2026
Mentored Teaching Fellowship, College of Engineering NC State, $1,000 June 2026
Carla Savage Award Nominee, Outstanding CS PhD Student NC State March 2026
Foundations of Software Engineering (FSE) Travel Grant, NSF, $2,200 June 2025

Grad Cohort Workshop Travel Grant, Computing Research Association, $480 October 2024

MEDIA COVERAGE

How Hannah Estes Found her Passion for Research, *NC State Department of Computer Science* October 2025

REVIEWER AND SUBREVIEWER

IEEE/ACM International Conference on Software Engineering (ICSE 2026): Software Engineering Education | 1 full paper

HCI International (HCII 2025) Workshop: Designing for Inclusivity: Advancing User Experience Through Inclusive Design Methodologies | 2 full papers